



P R E S E N T A T I O N

Houston Anodes

Protecting Offshore, Subsea and Onshore Assets and Infrastructure
since 1977.

Headquartered in the Energy Corridor of Houston, TX
100% Made in the USA

ISO 9001:2015

ISO 17025 Lab

DNV Certified

47+

YEARS

1,000+

PROJECTS

35 yr

NAVY SERVICE LIFE

0

CLAIMS



WHO WE ARE



Company Overview

Designing, Manufacturing and Delivering High-Quality Aluminum and Zinc Sacrificial Anodes. Equipped with a state-of-the-art laboratory and superior Electromagnetic Induction technology.

Over 1,000 Projects

Completed in the past five years across five continents.

Zero Claims

Not a single product claim in 47 years of operation.

U.S. Navy

Anodes with a proven 35-year service life.

TOP Energy Companies

Serving the world's Top 100 energy companies across five continents

OUR #1 PRIORITY

Safety Is Our #1 Priority

We commit to ensure no harm comes to our employees, or damage to the environment. The successful management of HSE is an essential part of our business model and corporate culture.



Employee Development

Ongoing training and development programs for technical skills and safety awareness at every level.



Safer Technology

Induction melting eliminates open-flame exposure and dangerous manual hand-stirring with steel rods.



Zero-Harm Culture

Prevention of all labor-related accidents. Active HSE participation from every stakeholder.

THE CHALLENGE

Corrosion Never Sleeps

Metals extracted from their primary ore have a natural tendency to revert to that state under the action of oxygen and water. Corrosion is an electro-chemical process that threatens safety, production and the environment.

\$50B+

Annual global losses

25%

Structures fail prematurely

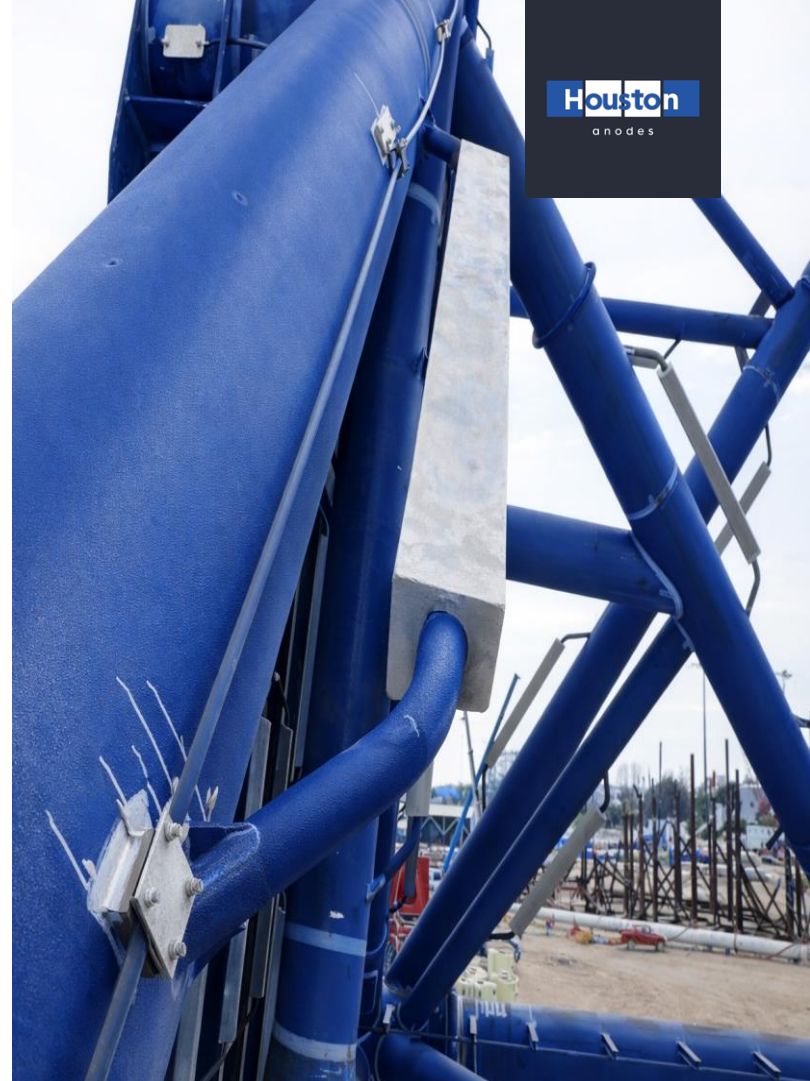
60%

Gulf platforms past 20-yr life

800 hrs

Avg. downtime / year

Houston
anodes



Cathodic Protection

The principle of Cathodic Protection is in connecting an external anode to the metal to be protected and the passing of an electrical DC current so that all areas of the metal surface become cathodic and therefore do not corrode.

- The current results from the potential difference between the two metals
- Causes a positive current to flow from the anode to the steel
- Surface of the steel becomes cathodic and does not corrode
- Requires: sacrificial anodes + direct welding to structure

Simple and Effective

Graphic / Chart Here



QUALITY IS OUR FOUNDATION

Quality Control

In-house state-of-the-art Laboratory equipped with a Bruker-Quantron Magellan Q8 spectrometer. Immediate, accurate test results.

- All raw materials inspected and tested upon arrival
- Chemical analysis performed on every heat
- Electrochemical properties tested per NACE TM-0190 on every batch
- Chemical results verified against MTRs
- Scheduled equipment calibration per OEM



Houston
anodes

ZERO CLAIMS

in 47 years of continuous operation

ISO 17025

Accredited laboratory

ISO 9001:2015

Certified by DNV

NACE TM-0190

Every batch tested

100% USA

All testing in-house

HIGHEST PURITY

Raw Materials



Houston Anodes fabricates anodes using only raw materials of the highest purity, sourced from premium U.S. and global suppliers.

99.9%

Primary Aluminum

99.99%

High Grade Zinc

100%

Indium

ASTM

Steel Cores

NEVER MELT SCRAP OR RECYCLED ANODIC MATERIAL

OUR MANUFACTURING ADVANTAGE

Houston
anodes

Electromagnetic Induction Technology

The furnace develops an electromagnetic field that creates eddy currents in the charge — generating heat without flames and without contact. Reliable, repeatable, extremely efficient.



Non-Contact Melting

100% homogeneous alloy. Consistent batches every time. Zero contamination.



50% Faster

Gas furnaces take 50% longer to melt and process than induction.



Enhanced Safety

No open flames. No manual stirring. Eliminates superheating risk.

725

TONS/MONTH

400kW

FURNACE 1

625kW

FURNACE 2

100% Made in the USA

WHY CONVENTIONAL METHODS FALL
SHORT

Gas Combustion Furnaces

Inherited Deficiencies

- Direct flame transfers hydrogen & carbon — contaminating the alloy
- Mixing by hand stirring with a steel rod — extremely dangerous
- Produces only heterogeneous mixtures — indium only partially induced
- Creates passive or inert portions, reducing anode lifespan
- **Takes 50% longer to melt and process**

[Gas Furnace Photo]

Induction vs. Gas Performance

All anodes installed at the same time. Performance at end of complete lifetime:

Gas Furnace (Competitors)

59%

ACTIVE CONSUMED

41%

PASSIVE WASTED

- Flame contamination & porosity
- Inconsistent quality batch to batch

Electromagnetic Induction (Houston Anodes)

100%

ACTIVE CONSUMED

0%

PASSIVE WASTED

- Zero porosity, perfect density
- **40% longer anode lifespan**

WHAT WE DELIVER



Product Portfolio

Aluminum alloy anodes from 5 lbs to custom configurations. We also manufacture our own molds.

Platform Anodes

90°/45°/straight core. Types A & B for welded jackets.
180–1,250 lbs.

Bracelet Anodes

10"–30" diameters. Square/tapered, weld-on/bolt-on.
22–265 lbs.

Pier, Piling & Ballast

Flush, stand-off, rod core configurations. Ballast
protection systems.

Tank & Vessel

Heater treater, eyebolt, bowling ball, drum. Internal
vessel protection.

Hull Anodes

Zinc hull for marine equipment. U.S. Navy: 35-year
proven service life.

Custom Anodes

Clamp, mat, link, wind turbines. Any geometry to exact
specification.

Light: 5–99 lbs

Medium: 100–400 lbs

Heavy: 401–1,650 lbs

Custom

Certifications & Compliance

Homogeneous alloys always conforming to exact international quality standards and specifications.

Standard	Description
ISO 9001:2015 (DNV)	Quality management system
ISO 17025	Laboratory accreditation
NACE SP-0387 / SP-0492	Sacrificial anode standards
DNV RP B401	CP design for offshore
ISO 15589-2:2012	Petroleum & gas industries CP
NORSOK M-503	Norwegian petroleum CP standard
NACE TM-0190	Electrochemical testing — every batch

WHO WE SERVE

Houston
anodes

Markets & Applications



Oil & Gas

Deepwater platforms, subsea structures, pipelines, refineries, chemical plants



Offshore Wind

Turbine foundations, monopiles. Delivered for Ocean Wind 1 & Revolution Wind



Infrastructure

Piers, pilings, harbor facilities, ballast systems, marine structures



Industrials

Heater treaters, tanks, vessels, drums, industrial corrosion protection



Defense — U.S. Navy

Hull anodes and marine equipment with a proven 35-year service life. Active defense-sector approvals.

GLOBAL TRUST, PROVEN RESULTS



Success Stories

Houston Anodes currently protecting major Offshore Assets and subsea structures throughout the world for valued clients including among others:

ExxonMobil	Shell	BP	Chevron	PEMEX	TechnipFMC
Subsea 7	J. Ray McDermott	Hess	Marathon	Williams	Anadarko
Oceaneering	Bredero Shaw	FloaTEC	Oil States	Sapura Energy	Ørsted
Deepwater	Noble	ATP Oil & Gas	LLOG	Corrpro	U.S. Navy

80%

PEMEX ANODES · 35+ YRS

35 yr

U.S. NAVY SERVICE LIFE

2

U.S. WIND PROJECTS

FMC

GLOBAL SUPPLIER

Houston Anodes

Protecting the Infrastructure
That Powers the World

47

YEARS

1,000+

PROJECTS

0

CLAIMS

100% Made in the USA

GET IN TOUCH

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ISO 9001:2015

ISO 17025

DNV

NACE

